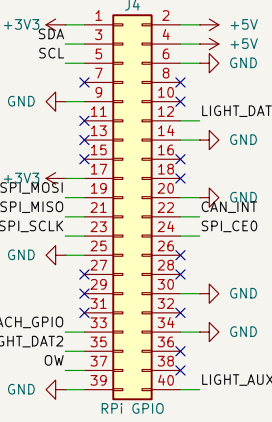
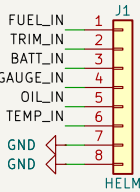


RASPBERRY PI HEADER



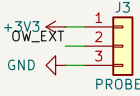
HELM: gauge taps and battery



TACH: Delco EST (Ignition side)



TEMPERATURE PROBES



Each gauge tap: the S (or I) terminal -> 10k in heatshink AT THE GAUGE -> J1 -> 39k -> node (10k to GND, BAT54S clamp to GND/3V3) -> 1k -> ADC, with 100n at the ADC pin. V_adc = V_tap x 10/59 (0-16 V -> 0-2.7 V). Battery: 47k on board, no remote resistor. x 10/57. Tach, Delco EST coil (BAT +12 V in, TACH out): J2-1 to the gray wire from the coil's TACH terminal, J2-2 to the tach gauge's G. The tach stays connected. TACH is the coil's switched side -- 12 V, ground while the coil charges, a few hundred volts at each spark: three 3.3k 1206 in series share the spike: 1W4148W and 10n across the LED. One rising edge per ignition cycle on GPIO13; 2 per revolution on the 3.0 4-cylinder (BOAT_TACH_PPR=2). Keep every TACH_* net (the ignition side) 3 mm from all other copper; see sensor-board.kicad_dru. Software: BOAT_SENSORS5=real BOAT_SENDER_WIRING=tap BOAT_TACH_PULL=none BOAT_TACH_GPIO=13 BOAT_OIL_SENDER=true BOAT_TEMP_SENDER=true; dtoverlay=w1-gpio.gpiopin=26. Channel map: fuel 0, trim 1, battery 2, gauge supply 3, oil 4, temp 5 -- see SENSOR_BOARD.md.

NMEA 2000: network side (isolated, powered by NET-S)

